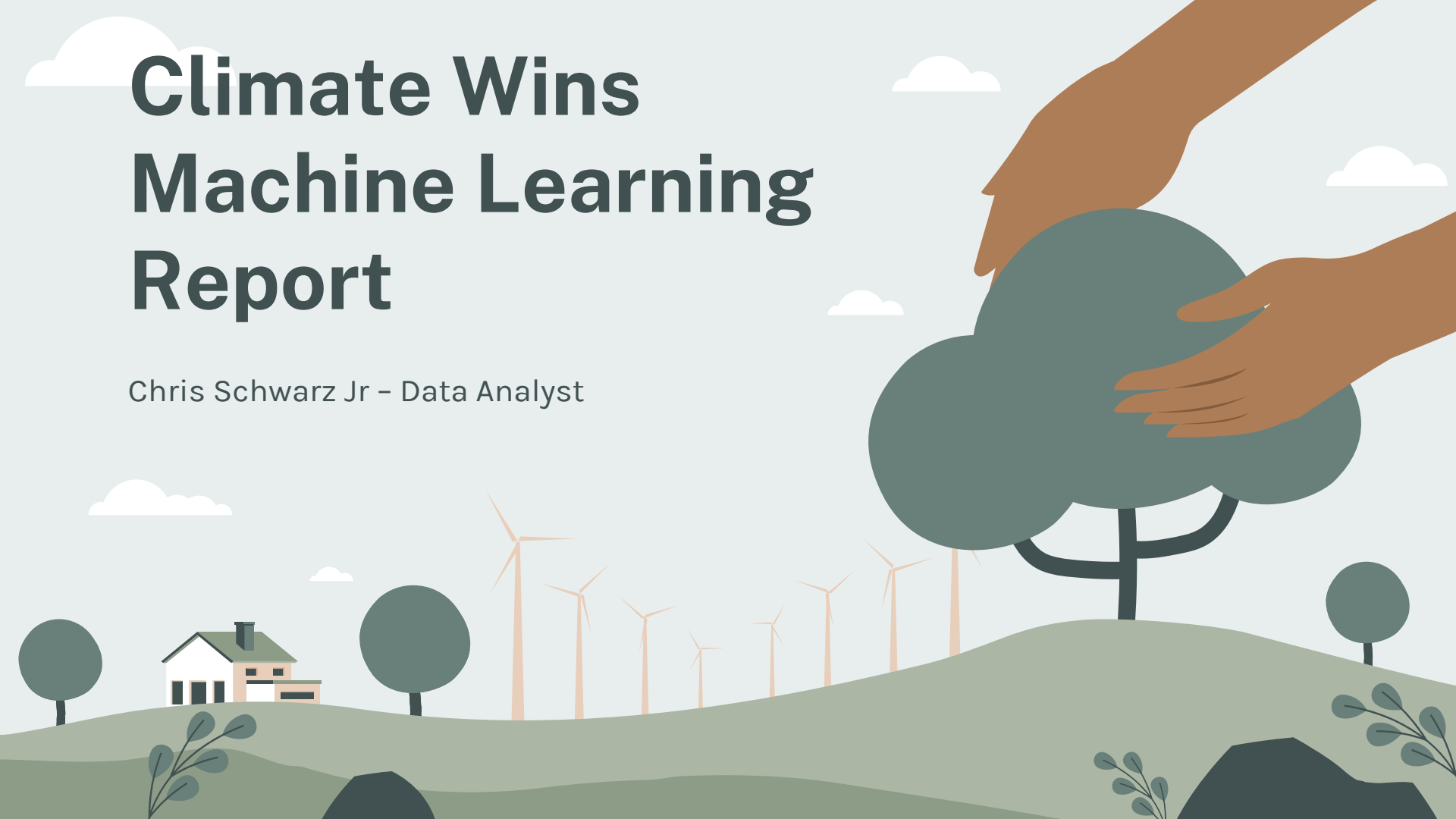


Climate Wins Machine Learning Report

Chris Schwarz Jr – Data Analyst

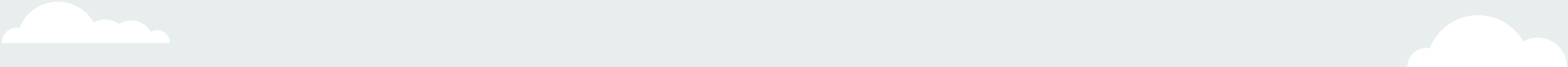


Background

As climate changes intensify, weather patterns worldwide have become increasingly unpredictable. Climate change poses significant threats to many communities across the globe.

ClimateWins aims to use machine learning to identify abnormal weather patterns, determine changes in weather over time, generate predictions for future weather patterns, and determine safe places for people to live.





Thought Experiments

01 Predicting Weather Events

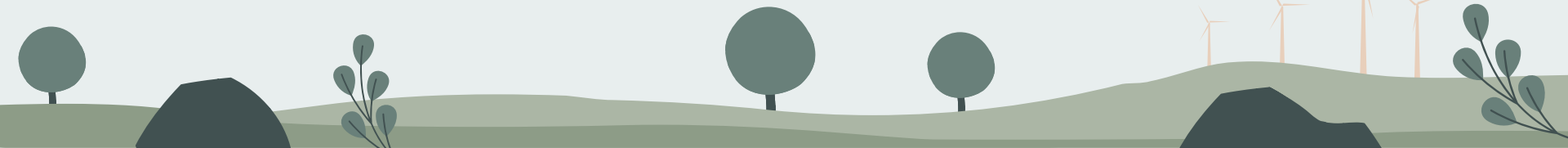
Using CNNs to predict unusual short term events that are harder to identify

02 Determining Safe Places to Live

Using Random Forests to classify regions that are safe from climate change

03 Non-Weather Climate Change

Using GANs & CNNs to predict fires and rising sea levels



Machine Learning Models



CNNs

- Convolution neural network
- Recognizes patterns
- Can process images such as radar
- Can be used to predict extreme weather events



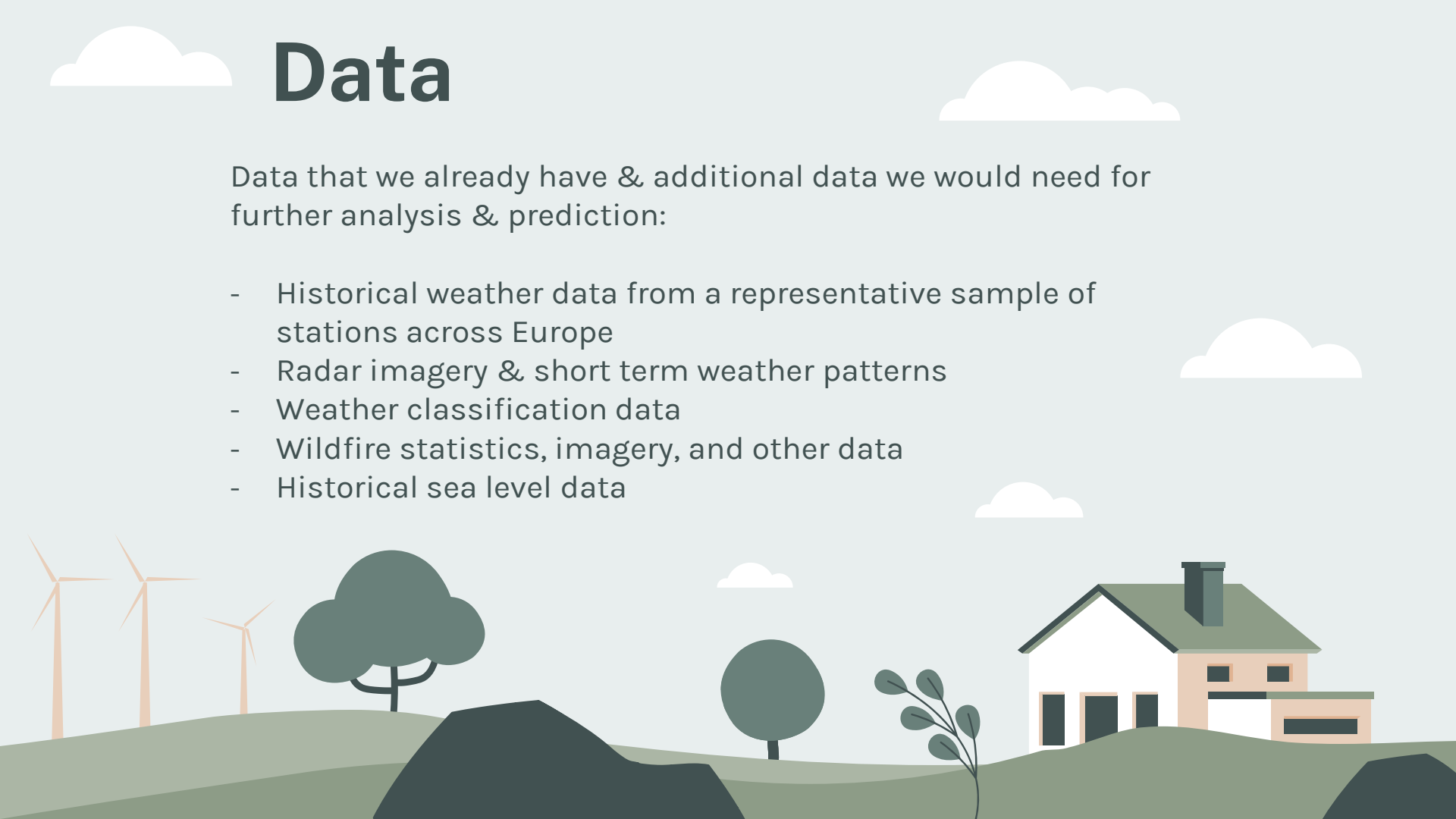
Random Forest

- Combines multiple decision trees using random subsets of data
- Classifies data
- Can be used to predict weather events
- Can be used to determine safe places to live



GANs

- Generative Adversarial Network
- Two network generator and discriminator competing to become more accurate
- Can create realistic artificial data to improve modeling

A stylized landscape illustration at the bottom of the slide. It features rolling green hills, a large dark grey rock in the foreground, several trees (one large and bushy, one smaller and round, and a small leafy branch), a white house with a green roof and a chimney, and three wind turbines on the left. The sky is light blue with several white clouds.

Data

Data that we already have & additional data we would need for further analysis & prediction:

- Historical weather data from a representative sample of stations across Europe
- Radar imagery & short term weather patterns
- Weather classification data
- Wildfire statistics, imagery, and other data
- Historical sea level data

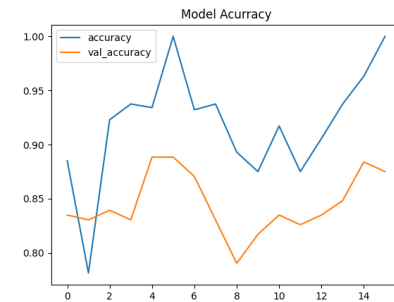
Thought Experiment #1: Predicting Weather Events

Problem: Short term weather events can be harder to predict than long term changes.

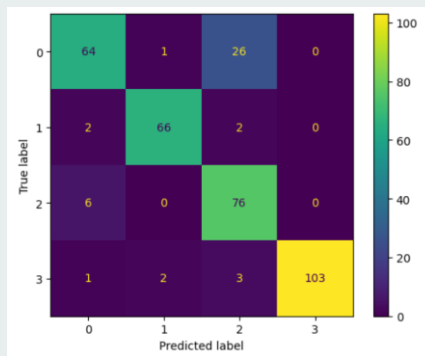
Hypothesis: CNNs can aid in the prediction of short-term extreme weather events

Approach: Optimize a CNN model using radar and photographic imagery so that the model can identify future weather patterns

Conclusion: If a CNN can make accurate predictions based on radar and other imagery, the relevant authorities can quickly take necessary action to protect people, property & infrastructure



Graph displaying the accuracy of a model of the course of 16 epochs



Confusion matrix showing correct & incorrect predictions

Thought Experiment #2:

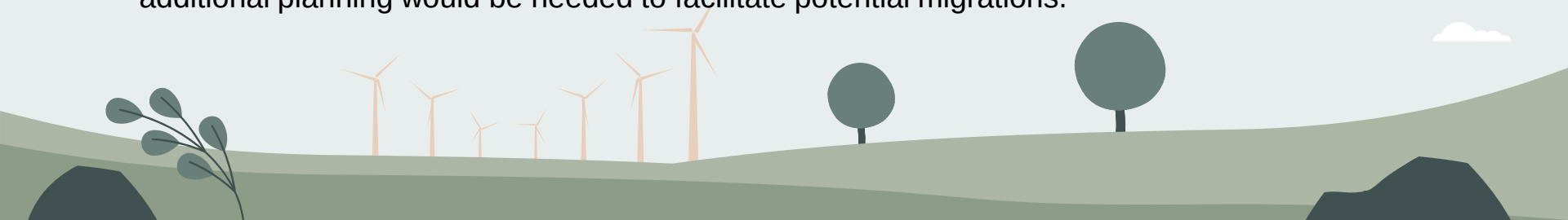
Determining Safe Places to Live

Problem: Climate change will inevitably impact the habitability of numerous locations worldwide. The safety of many people will be impacted by climate change and may necessitate migration.

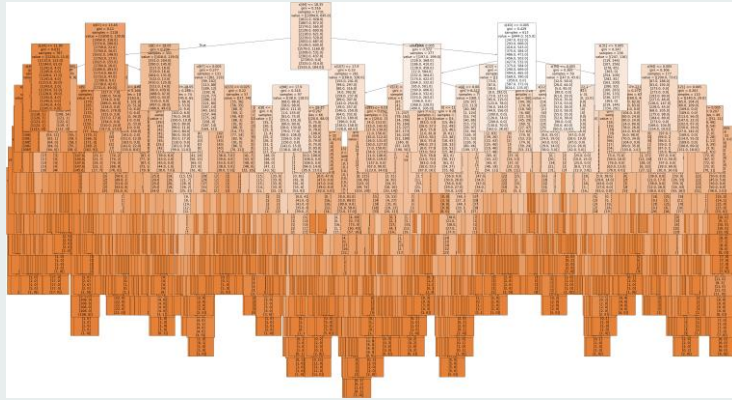
Hypothesis: Random Forests can help classify locations based on their longtime viability given the risks of climate change

Approach: Create a Random Forest model that can classify locations based on climate change predictions. Prune the forest by potentially splitting the data into subsets and optimize based on regional differences

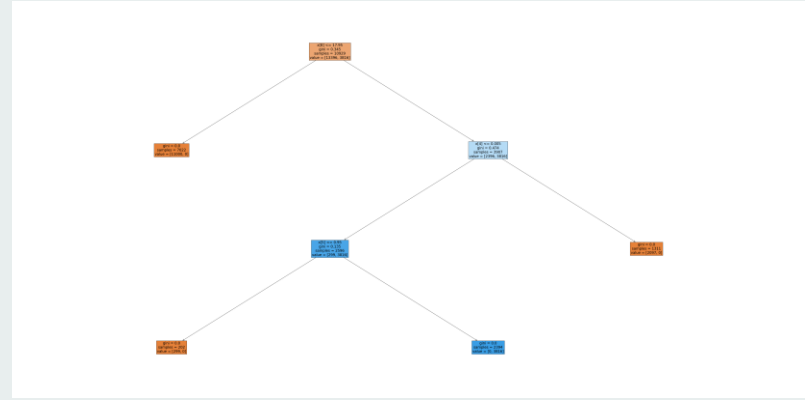
Conclusion: An optimized random forest could classify cities and regions based on their long-term viability. This could help individuals and authorities when planning for the future. However, much additional planning would be needed to facilitate potential migrations.



Thought Experiment #2: Determining Safe Places to Live



This random forest has too many branches to effectively interpret. However, it does show the complexity that random forests can manage.



This random forest has been pruned to only a few branches, making it easier to read. However, it only applies to a single weather station.

Thought Experiment #3:

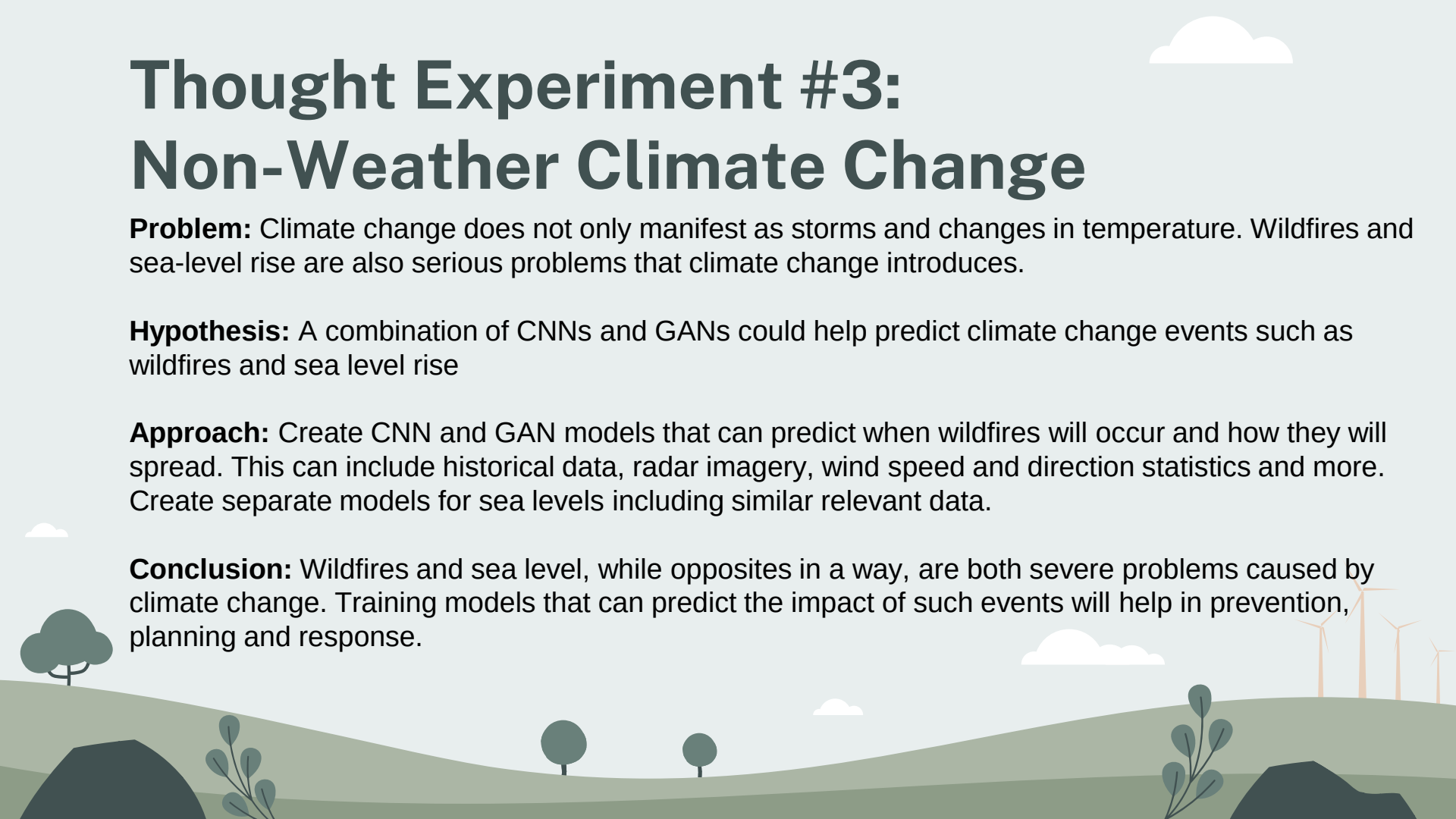
Non-Weather Climate Change

Problem: Climate change does not only manifest as storms and changes in temperature. Wildfires and sea-level rise are also serious problems that climate change introduces.

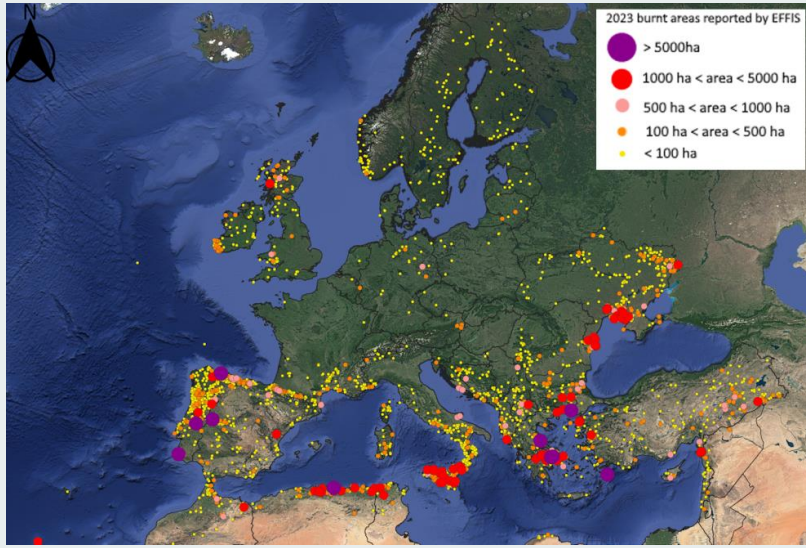
Hypothesis: A combination of CNNs and GANs could help predict climate change events such as wildfires and sea level rise

Approach: Create CNN and GAN models that can predict when wildfires will occur and how they will spread. This can include historical data, radar imagery, wind speed and direction statistics and more. Create separate models for sea levels including similar relevant data.

Conclusion: Wildfires and sea level, while opposites in a way, are both severe problems caused by climate change. Training models that can predict the impact of such events will help in prevention, planning and response.



Thought Experiment #3:



Map of Wildfires in Europe. In 2023 Wildfires emitted 20 million tons of carbon dioxide into the atmosphere.

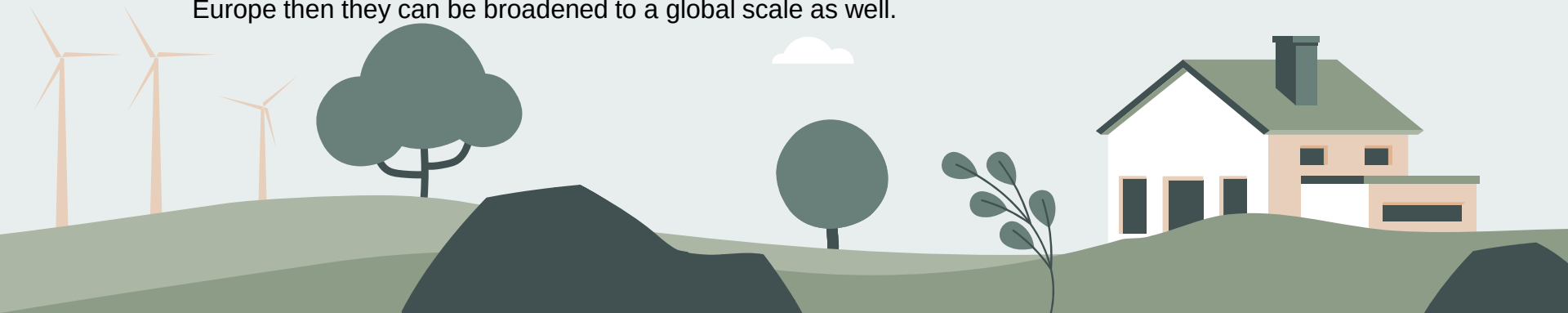
Source



Regions of Europe at risk as water levels rise. Major cities such as Amsterdam & Venice face substantial risks as water levels rise. Source

Summary

- **Thought Experiment #1** could be the simplest to enact and would be impactful in the short-run. However, it would require a more robust data set of additional weather stations and radar imagery, alongside a more powerful algorithm that can make short term assessments.
- **Thought Experiment #2** can assess the problem of habitability; however, it doesn't solve the problem itself. As cities become increasingly difficult to live in there could be a corresponding mass migration to places less affected by climate change. So, significant additional planning and infrastructure would be needed to correspond with this model for it to be impactful. A powerful machine learning model could aid in this process.
- **Thought Experiment #3** addresses some additional climate related issues. While Europe doesn't suffer from climate change or sea-level rise as severely as other parts of the world, if these models could be optimized for Europe then they can be broadened to a global scale as well.



Thanks!

Do you have any questions?

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